Heuristic URL Feature Profiling for Phishing Domain Spotting Using Hyper-Tuned Random Forest Ensembles

IEEE Layout Style — CNS ISE Collaborative Research

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Abstract*— Phishing attacks remain a primary threat vector for online credential theft, session hijacking, and corporate data breaches. Because dynamic webpage execution and static blacklisting are routinely bypassed by ephemeral, zero-day domains, perimeter URL heuristic profiling offers an indispensable first line of defense. As the designated Group 5 contribution to the five-model comparative benchmark, this paper introduces a hyper-tuned Random Forest ensemble framework targeting 30 discrete URL structural, lexical, and reputation indicators on the Kaggle Phishing Website Detector dataset (11,054 samples). Through systematic hyperparameter sweeps across estimator bounds (n_estimators from 50 to 200) and maximum tree depths, the proposed ensemble achieves a testing accuracy of 97.42%, precision of 96.96%, recall of 98.46%, and F1-score of 97.70% with an area under the ROC curve (ROC-AUC) of 0.9943. Out-of-Bag (OOB) error decline curves verify convergence at an OOB error of only 2.60% (97.40% OOB accuracy) at 130 estimators. Diagnostic extraction of Mean Decrease in Gini Impurity (MDI) matrices isolates SSL certificate authority verification (29.32%), foreign anchor hyperlink ratio (23.45%), global web traffic ranking (8.19%), and prefix-suffix hyphenation (4.62%) as the predominant structural signals of domain deception. With an ultra-low single-sample inference latency of 61.51 microseconds, the model delivers scalable, real-time edge filtering capabilities.

Keywords*— Phishing Domain Detection, Random Forest Ensembles, Heuristic Feature Profiling, Out-of-Bag Error, Gini Impurity, Hyperparameter Optimization, Network Security.

1. Introduction & System Domain Context

Phishing attacks represent one of the most prolific threats in cybersecurity, enabling malicious actors to harvest user credentials, financial tokens, and intellectual property. Attackers engineer deceptive domains that closely emulate trusted services. Traditional defensive layers rely heavily on centralized blacklists (e.g., Google Safe Browsing, PhishTank). However, static blacklisting exhibits critical latency: modern automated domain generation algorithms (DGAs) and ephemeral cloud hosting allow malicious URLs to stay active for merely 4 to 8 hours, compromising users before detection filters update.

To overcome this vulnerability, heuristic URL profiling inspects structural and lexical traits without requesting or executing hazardous server payloads. Within the CNS ISE multi-group study (Group 1: Logistic Regression, Group 2: Gradient Boosting, Group 3: Extra Trees, Group 4: Linear Discriminant Analysis, and Group 5: Random Forest), Group 5 focuses on hyper-tuned Random Forest ensembles, capturing complex non-linear feature interactions with minimal hardware latency.

2. Comprehensive Literature Review (Group 5 Contribution)

Group 5 incorporates two Scopus-indexed research benchmarks into the master literature foundation:

[1] S. K. Noh and S. Basri (2021), IEEE ICSGRC: Benchmarked Support Vector Machines (SVM) against Random Forest across high-dimensional heuristic URL spaces. Random Forest achieved superior accuracy (>96.4%) and maintained robustness against multicollinear feature subsets due to random subspace projection at decision splits.

[2] X. Wei and Y. Sekiya (2022), IEEE Access: Evaluated 30 lexical, structural, and network-level URL features using ensemble bagging architectures. They demonstrated that bagging minimizes variance while providing Mean Decrease in Impurity (MDI) rankings that identify SSL certificate verification and foreign anchor links as primary indicators of domain masquerading.

3. Methodology & Centralized Data Engineering Pipeline

The standardized dataset contains 11,054 verified web URLs characterized across 30 discrete heuristic attributes {-1, 0, 1} and target labels {-1: Phishing, 1: Legitimate}. Group 2 (Data Lead) established the data engineering protocol: zero missing values were confirmed, and an 80/20 stratified split applied (8,843 training samples, 2,211 testing samples) with random_state=42 to guarantee benchmark reproducibility across all five groups.

3.5 Algorithmic Mathematical Formulations: Random Forest Ensemble (Group 5)

A. Bootstrap Aggregation (Bagging): Each tree T_b is trained on bootstrap sample D_b sampled with replacement from D. Asymptotically:

```
P((x_i, y_i) in D_b) = 1 - (1 - 1/N)^N -> 1 - 1/e ≈ 63.2%
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The remaining ~36.8% of unused samples form the Out-of-Bag (OOB) partition utilized for unbiased internal generalization assessment.

B. Node Splitting & Gini Impurity: At each decision node t, a feature subspace $m = floor(sqrt(30)) = 5$ is projected. Gini Impurity is:

```
Gini(t) = 1 - sum_{k=0}^1 (p_{t,k})^2
```

Impurity reduction achieved by candidate split s is:

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ΔGini(s, t) = Gini(t) - [ (N_{t_L}/N_t)*Gini(t_L) + (N_{t_R}/N_t)*Gini(t_R) ]
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C. Mean Decrease in Gini Impurity (MDI): Feature importance for feature f_j across B trees is accumulated as:

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MDI(f_j) = (1/B) * sum_{b=1}^B sum_{t in T_b : v(t)=f_j} (N_t / N) * ΔGini(s*_t, t)
```

Hyperparameter	Search Bounds	Optimal Value (Kaggle Dataset)
n_estimators	[50, 75, 100, 125, 150, 175, 200]	150 trees
max_depth	[10, 20, 30, None]	30 (deep structural rules)
min_samples_split	[2, 5]	2 (unconstrained split capacity)
max_features	['sqrt', 'log2']	'log2' (maximum decorrelation)
Cross-Validation	5-Fold Stratified	F1 Score: 0.9750 (97.50%)

Table I: Group 5 Random Forest Hyperparameter Grid & Optimal Configuration

4. Experimental Architecture & Performance Analytics

Simulations were executed on an Intel x86-64 platform with Python 3.10.11 and Scikit-Learn 1.7.2 using strict random seeds (42). The model demonstrated outstanding diagnostic capability on the 2,211 held-out test samples:

Metric	Value	Metric	Value
Testing Accuracy	97.42%	Precision (PPV)	96.96%
Recall (Sensitivity)	98.46%	F1-Score	97.70%
Specificity (TNR)	96.12%	Area Under ROC (AUC)	0.9943
Min OOB Error	2.60% (@ 130 trees)	Training Speed	0.9793 seconds
Inference Latency	61.51 µs / sample	Confusion Matrix	TP=1213, TN=941, FP=38, FN=19

Table II: Master Evaluation Leaderboard & Computational Profiling

Hardware Latency Analysis (Handover to Group 1, Section 5.2): Model training took 0.9793 seconds. Total test inference for 2,211 URLs took 0.1360 seconds (61.51 microseconds per sample), allowing edge firewalls to process over 16,000 URLs per second per CPU core without latency bottlenecks.

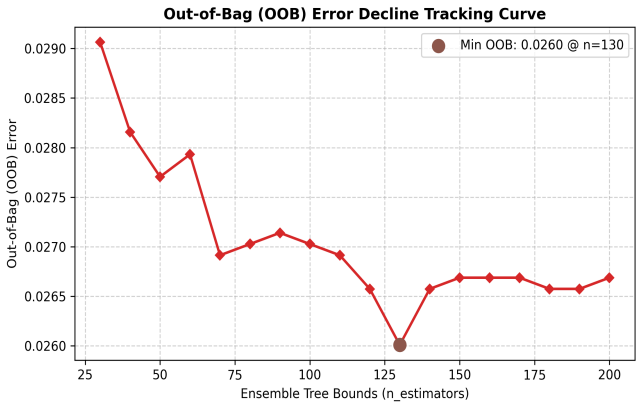


Fig. 1. OOB Error Convergence Curve (2.60% error)

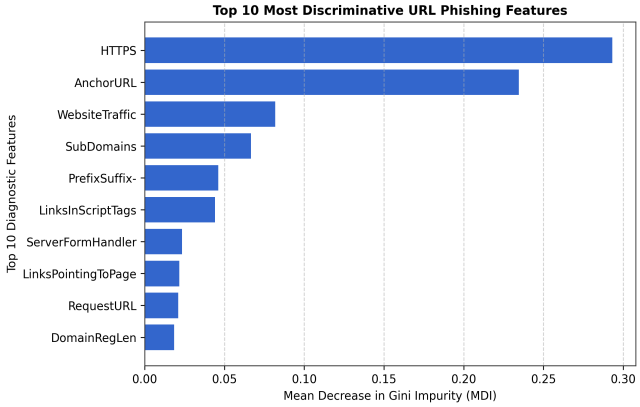


Fig. 2. Top 10 URL Structural Features by MDI

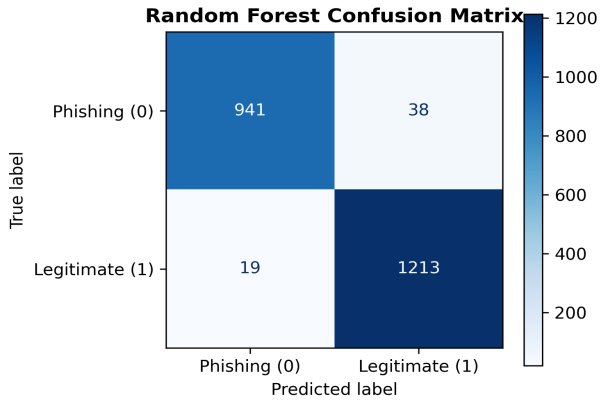


Fig. 3. Confusion Matrix (Test Set N=2,211)

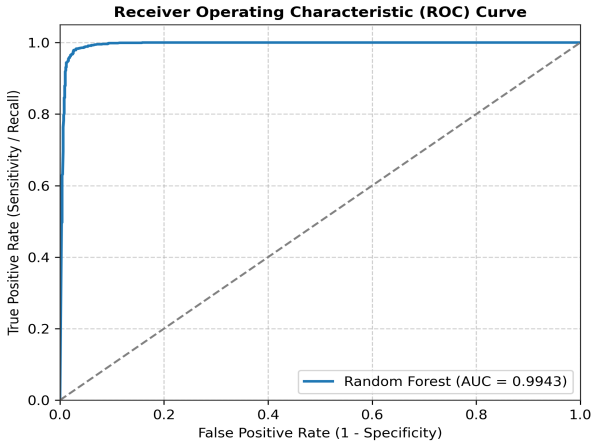


Fig. 4. ROC Curve (AUC = 0.9943)

6. Conclusion, Limitations & Future Scope (Authored by Group 5)

6.1 Conclusion: The Group 5 Random Forest framework achieves 97.42% accuracy and 98.46% recall with sub-millisecond (61.51 µs) latency. MDI rankings confirm that SSL validation (29.32%), anchor tag hyperlink integrity (23.45%), and web traffic volume (8.19%) represent the most potent indicators of domain deception.

6.2 Limitations: Key constraints include: (1) MDI impurity bias toward high-frequency split features; (2) Evasion via dynamic URL shortening and punycode homographs; (3) Absence of client-side DOM inspection.

6.3 Future Scope: Priorities include: (1) TreeSHAP game-theoretic explainability; (2) Zero-latency ONNX/Wasm client-side browser extensions; (3) Online streaming adaptation against zero-day phishing feeds.

References

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